

THREADLINE / REVERIE HACKS 2026

Why every workflow node exists.

Human inputs, five model operations, deterministic validation, retrieval, contradiction and rival analysis, release policy, human review, audit, replay, limits, and reproduction.

SCOPE

Possible record connections for authorized human review only. Synthetic evidence; no autonomous identity confirmation and no operational humanitarian validation.

Status: synthetic research demonstration Workflow version: threadline-workflow/2.0.0 Production extraction prompt: extraction/v2
Decision boundary: possible record connections for authorized human review only

RESEARCH BOUNDARY

AI extracts source-cited evidence. Deterministic comparison and conflict policies decide when the system must stop.

1. Research claim

THREADLINE tests whether a decomposed, source-cited prompt workflow can make record-reconciliation evidence safer to review than a one-call answer. The system preserves source lineage, asks separate model operations to build and challenge a hypothesis, retains rival candidates, and applies deterministic release rules around every probabilistic output.

It does not claim that a prompt proves identity. It does not autonomously merge records, declare someone found, or replace authorized humanitarian review.

The ML prompt-engineering contribution is the workflow boundary:

- constrain each model operation to one reviewable purpose;
- require schema-constrained output rather than prose;
- bind extracted claims to exact source spans;
- keep supportive and adversarial reasoning separate;
- prevent a model operation from upgrading a deterministic block; and
- evaluate prompt iterations on extraction precision, recall, false-positive fields, downstream safety states, and reproducibility - not completion rate alone.

2. End-to-end workflow

The editable source is reverie-ml-workflow.svg. The implementation contains twelve ordered nodes, grouped into seven judge-readable stages.

Order	Runtime node	Type	Purpose	Input	Output	Prompt / tool	Deterministic boundary	If missing or invalid	Why it is separate
1	incident	Human + deterministic	Establish incident scope, fictional-data declaration, limits, and review boundary.	AnalyzeRequest	IncidentConfigurationOutput	Pydantic validation	Record count, text size, and candidate limits	Reject request	Keeps authorization and scope outside model inference.
2	quarantine	Deterministic	Treat every narrative as untrusted text and isolate tested instruction-like patterns.	RecordInput[]	QuarantinedRecord[]	Quarantine service	Original text must remain unchanged; safe rendering is separate	Fail before provider call	Prevents record text from becoming workflow instruction.
3	extract	LLM	Convert narrative evidence into typed, cited fields.	QuarantinedRecord[]	ExtractionModelOutput[]	extraction/v2	Schema, canonical-key allowlist, exact quote/offset validation	Unsupported field is withheld; one bounded repair attempt	Separates probabilistic reading from evidentiary admissibility.
4	normalize	LLM + deterministic helpers	Add multilingual and canonical comparison views without replacing originals.	Validated extractions	NormalizationModelOutput[]	normalization/v1 + Unicode/name helpers	Original form and evidence links must survive	Reject representation that loses lineage	Keeps representation distinct from identity conclusion.
5	timeline	Deterministic	Reconstruct order while preserving exact versus estimated status.	Normalized evidence	TimelineEvent[]	Timestamp and chronology rules	Approximation cannot become fact	Withhold coherent timeline if it requires invention	Makes time/location contradictions inspectable.
6	retrieve	Retrieval + deterministic	Generate bounded candidates and compare fields.	Normalized records	CandidateConnection[]	Blocking, top-k retrieval, pairwise comparison, scoring	Missing = agreement; score = probability; hard conflicts block	insufficient_evidence when discriminating evidence is absent	Separates search recall from release policy.
7	hypothesis	LLM	Summarize the strongest admissible support for one candidate.	Candidate evidence	HypothesisModelOutput	hypothesis/v1	Every factor must cite a valid source span; blocked states cannot upgrade	Remove unsupported support	Prevents the affirmative case from absorbing opposition.
8	prosecutor	LLM	Independently search for contradiction and missing support.	Candidate evidence	ProsecutorModelOutput	prosecutor/v1	Deterministic blockers are restored if omitted; absence is not conflict	Unresolved material conflict continues to the gate	Makes disconfirming review an explicit prompt operation.

Order	Runtime node	Type	Purpose	Input	Output	Prompt / tool	Deterministic boundary	If missing or invalid	Why it is separate
9	rivals	Retrieval	Test whether evidence distinguishes the leader from nearby alternatives.	Candidate set	RivalComparison[]	Deterministic rival comparison	Close rivals remain visible; shared contacts are not independent proof	Close rival can force insufficiency/review	A leading candidate is unsafe without alternatives.
10	adjudicate	LLM + deterministic guard	Aggregate structured support, opposition, and rivals.	Adjudication Packet	AdjudicationModelOutput[]	adjudication/v1	Transition allowlist permits preserve or downgrade only	Retain safer deterministic state	Allows concise review context without surrendering policy control.
11	privacy	Deterministic	Minimize the reviewer-facing packet while retaining audit evidence.	Candidate review packet	PrivacySafePacket	Pattern redaction	Restricted configured fields cannot remain in safe_text	Fail packet	Makes reviewer minimization explicit and testable.
12	review	Human routing	Prepare an authorized-review packet and permitted actions.	Privacy-safe packet	HumanReviewerRoute	Deterministic router + human disposition	No autonomous identity-confirmation action	Packet invalid if it bypasses review	Separates software preparation from accountable human action.

After node 12, an 18-rule deterministic evidence contract evaluates source integrity, lineage, certainty preservation, hard-conflict inclusion, rival coverage, safety language, audit integrity, and release authorization. Any blocking outcome clears the externally releasable classification and marks the packet Withheld.

3. The five model operations

The runtime selects provider and model using LLM_PROVIDER and LLM_MODEL; that identity is recorded in run metadata. Offline Judge Mode uses a deterministic mock provider and is labelled not model performance. The archived live Prompt V2 artifact evaluates extraction with an OpenAI-compatible provider and Qwen/Qwen3-30B-A3B-Instruct-2507 at temperature 0; it does not prove that all five nodes ran live under that model.

#	Model operation	Prompt and SHA-256	Query purpose	Expected structured output	Validation boundary	Evaluation signal
1	Structured extraction	extraction/v2 92302c609100a882463cb950dd315f0185541082d030e7db622e570ce04b2624	Extract every explicitly supported identity-relevant field while preserving raw script, partial precision, and exact spans.	ExtractionModelOutput[]	Pydantic, canonical keys, quote equality, half-open offsets, certainty enum	TP/FP/FN, micro precision/recall/F1, extraction success, per-field error analysis
2	Multilingual normalization	normalization/v1 7b4ad0c542bb02470161ef262e7661f248b33a19881b50f9adca20a540a91c4e	Propose comparison forms while keeping source-language values authoritative.	NormalizationModelOutput[]	Original form and evidence links preserved; deterministic canonicalizers retained	Cross-script retrieval and candidate behavior; lineage validity

#	Model operation	Prompt and SHA-256	Query purpose	Expected structured output	Validation boundary	Evaluation signal
3	Evidence hypothesis	hypothesis/v1 25c04d5abd5cc97680a239d00d29c159c3ec49bed4edd0787b96313eb5df234b	State only the strongest source-backed support for a candidate pair.	HypothesisModelOutput	Every factor cites an admissible span; unsupported support removed	Exact-citation validity; unsupported factor count; downstream state cannot upgrade
4	Contradiction challenge	prosecutor/v1 a9091ce340e13e7028a150c682e60dd45e4ae6d67d3ba1d0e2dc32e5bcf5b2f7	Search separately for conflicting fields, timeline impossibility, and insufficient independence.	ProsecutorModelOutput	Deterministic blockers restored; hard/soft conflict types retained	Blocking-conflict recall; contradiction recall; unsupported opposition
5	Independent adjudication	adjudication/v1 a88a3e2ae0b6b1bcaf81e58176d42b26995a1d625a34d39dfe3ee95b888048c	Produce concise review context from isolated support, opposition, and rival packets.	AdjudicationModelOutput[]	Explicit transition allowlist: preserve or downgrade, never upgrade blocked/insufficient	Invalid transition count; disposition consistency; contract invalid-release rate

These outputs are structured decision rationales, not private chain-of-thought.

4. Prompt iteration and promotion policy

Prompt versions are promoted by measured evidence, not recency.

- V1 established exact-span extraction and canonical keys.
- V2 / PRODUCTION expanded completeness, multilingual preservation, OCR tolerance, distinguishing-mark handling, partial-value preservation, and a final extraction checklist.
- V3 / NOT PROMOTED added stricter subject-attachment and field-specific precision instructions. On the full 58-record archive it completed 58/58 extractions and improved recall, but emitted substantially more false-positive fields, reducing precision and F1. Completion alone was not accepted as progress.

The apples-to-apples comparisons are intentionally separated:

Shared cohort	Prompt	Success	TP / FP / FN	Precision	Recall	F1
Targeted 19 records	V1	19/19	56 / 4 / 9	0.9333	0.8615	0.8960
Targeted 19 records	V2	19/19	58 / 2 / 7	0.9667	0.8923	0.9280
Full 58 records	V2	57/58	157 / 30 / 25	0.8396	0.8626	0.8509
Full 58 records	V3	58/58	161 / 78 / 21	0.6736	0.8846	0.7648

On the full cohort, V3 added 4 true-positive fields and recovered 4 previously missed fields, but also added 48 false-positive fields relative to V2. Production remains V2 because the precision regression is unsafe for an evidence workflow.

5. Evaluation design

THREADLINE keeps two evidence tracks separate.

Track A - deterministic same-input workflow replay

- 21 fixed synthetic cases: 12 same-identity, 8 different-identity, 1 deliberately ambiguous;
- fixed seed 41027;
- exact/fuzzy, generic one-prompt, structured one-call, and full workflow systems;
- identical record packets per compared system; and
- deterministic mock outputs labelled not model performance.

This track demonstrates workflow behavior and reproducibility. It does not measure live LLM quality.

Track B - archived live-provider extraction

- 58 synthetic records;
- OpenAI-compatible provider;
- model Qwen/Qwen3-30B-A3B-Instruct-2507;
- temperature 0;
- fixture SHA-256 eed1032519f78ebfd18a49da55c920c1b280ade1b7e32ebda514157ff1ead850;
- versioned prompt and raw-output artifacts; and
- deterministic downstream retrieval and policy evaluation.

This is one archived run per full prompt version, not a repeated-trial confidence study and not a same-model one-shot versus workflow experiment.

Track C - measured live same-model one-shot versus workflow comparison (frozen v1.1, completed)

A same-model comparison was preregistered before any provider output was observed as spec THREADLINE-REVERIE-LIVE-EVAL-V1.1 (SHA-256 458279a9595b79e3785af7004cf68a93223fde21e0934b4c432525bebd786a1). The draft v1 spec (dccfb171...1c99) is preserved as superseded-before-observation. Design: 3 synthetic cases 3 repetitions 2 systems = 18 top-level runs; both systems receive identical records, incident context, provider (openai_compatible), model (Qwen/Qwen3-30B-A3B-Instruct-2507), and temperature 0. Every provider request, response, retry, and error is recorded in a sanitized, resume-safe checkpoint. The run completed 18 / 18 top-level runs with 0 final failures; three recorded schema-validation failures were retried to success under the symmetric policy and remain visible in the recorder.

System	Schema-valid	Outcomes within accepted set	Invalid citations	Supported / unsupported fields	Provider calls	Transport attempts	Tokens
Structured one-shot	9 / 9	9 / 9	45	9 / 45	9	18	20,108
Full THREADLINE workflow	9 / 9	3 / 9	0	3 / 0	82	142	150,875

Measured behavior is mixed and is reported without spin:

- The one-shot returned an acceptable classification in all 9 runs, but 45 of its 54 emitted claim spans were inexact (exact-citation validity 9/54), including 7 invalid citations in every cross-script run.
- THREADLINE emitted zero invalid citations and zero unsupported fields, but returned insufficient_evidence in all 9 runs - including all 3 cross-script runs where the frozen ground truth accepts only positive or human-review outcomes, and all 3 blocking-conflict runs where the expected state is blocked_by_conflict. On these live runs it under-detected the DOB conflict rather than surfacing it.
- This is a small synthetic experiment (3 cases 3 repetitions). It establishes no statistical significance, no system superiority, and no operational validity.

Authoritative artifacts: backend/data/reverie_live_evaluation_v1_1/results.json, the sanitized recorder raw-provider-calls.json, the byte-identical submission copy docs/submission/reverie-live-evaluation-results.json, and the publication manifest docs/submission/reverie-live-evaluation-manifest.json.

6. Representative safety case

The judge case uses the exact same three-record packet for each deterministic comparison system:

- FAMILY-018 - family report, age 14;
- SHELTER-204 - shelter intake, age 14; and
- HOSPITAL-052 - plausible rival, documented age 24 from an identity card.

Input SHA-256: bc8084e71ab87554163279a3b61d27d1b71f0c06fb3347c4555de5670ab83576 Complete available-evidence
SHA-256: 68c34760bc4c0f0f6e18f0aeb40d1ddf5243656efcc75b7034dbe0d73223ff20

The generic one-prompt replay returns FAMILY-018 <-> SHELTER-204 as a possible candidate with no citations, contradiction, or rival comparison. THREADLINE preserves that pair as a possible connection for authorized review while blocking candidate pairs involving HOSPITAL-052 under TIMELINE_CONSISTENCY and MATERIAL_CONTRADICTIONS_RESOLVED, with exact age spans attached. This is not an identity confirmation.

7. Audit and reproducibility

Run artifacts preserve, where available:

- case and record IDs;
- exact source text and source-span coordinates;
- prompt/template ID, version, and content hash;
- provider, model, temperature, run timestamp, and run ID;
- fixture and identity-assignment hashes;
- raw model outputs, validation failures, retries, and repair attempts;
- deterministic comparison and evidence-contract outcomes; and
- audit, replay, and counterfactual records.

Authoritative sources:

- submission/results.json
- submission/results.csv
- submission/raw-outputs.json
- submission/ablation-counterfactual.json
- ../backend/data/live_runs_phase11/full-prompt-v2-run-1/live-artifact.json
- ../backend/data/live_runs_phase13/full-prompt-v3-run-1/live-artifact.json

8. Prior work and scope

THREADLINE does not claim to have invented digital missing-person matching. The ICRC Missing Persons Digital Matching project already describes multilingual comparison and human analysis of candidate results. THREADLINE's narrower experimental contribution is exact claim-to-source lineage, adversarial contradiction analysis, retained rivals, prompt-injection containment, deterministic release rules around probabilistic stages, replay/counterfactual evaluation, and safety-weighted synthetic benchmarking.

The design is informed by the ICRC Handbook on Data Protection in Humanitarian Action and the revised OCHA Data Responsibility Guidelines. THREADLINE is not endorsed by either organization and has not been validated for operational humanitarian use.

9. Known limitations

- All records and identities are fictional and synthetic.
- The deterministic baseline comparison is not live-model evidence.
- The live archives contain one full run per V2/V3 version and no repeated-run confidence intervals.
- The frozen live same-model comparison (3 cases 3 repetitions) completed 18/18, but its outcomes are mixed: the one-shot was outcome-acceptable in 9/9 runs with 45 invalid citations, while THREADLINE was citation-exact (0 invalid) but abstained in all 9 runs and under-detected the blocking DOB conflict. No superiority claim is made from this small synthetic experiment.

- Zero observed false merges uses only eight evaluated different-identity opportunities in the archived V2 run; it is not a field-safety guarantee.
- Selected workflow-node ablations produced no primary-metric or case-level change on the small fixed harness; that no-effect result is retained.
- Authentication, role enforcement, encryption at rest, automated retention/deletion, connected-provider exact replay, field validation, fairness evaluation, and external security review remain production requirements.